NATIONAL ENGINEERING COLLEGE, K.R.Nagar, Kovilpatti – 628 503.

(An Autonomous Institution - Affiliated to Anna University, Chennai)

Estd : 1984

Subject Code: 23IT51C	Subject Name: Cryptography And Network Security	
Degree & Branch	B.Tech & Information Technology	Semester: **V**
Regulations	2023	Question Pattern: **F**
Time	Three hours	Total Marks: 100
Name, Designation & Department of the Question Bank Setter(s)	Ms. M.Malathi, Assistant Professor/IT Ms.M.Akila, Assistant Professor/IT	
Course Outcome	**CO1:** Examine security principles and classical encryption techniques to assess and mitigate various threats.	

PART-A
(2 Marks)

Q.No.		Marks
1.	Distinguish between passive and active security attacks. Name some passive and active attacks.	(2)
2.	In the Permutation Cipher, suppose the permutation is using the key π(x) to decrypt the following ciphertext: "TGEEMNELNNTDROEOAAHDOETCSHAEIRLM" X: 1 2 3 4 5 6 7 8 π(x): 4 1 6 2 7 3 8 5	(2)
3.	Define the type of security attack in each of the following cases: i. A student breaks into the professor's office to obtain a copy of the next day's test. ii. A student gives a cheque for $10 to buy a used book. Later, she finds that the cheque was cashed for $100. iii. A student sends hundreds of emails per day to another student using a phony return email address.	(2)
4.	Which technique (cryptography or steganography) is used in each of the following cases for confidentiality? a. A student writes the answers to a test on a small piece of paper, rolls up the paper, and inserts it in a ball-point pen, and passes the pen to another student. b. To send a message, a spy replaces each character in the message with a symbol that was agreed upon in advance as the character's replacement. c. A company uses special ink on its checks to prevent forgeries. d. A graduate student uses watermarks to protect her thesis, which is posted on her website.	(2)

5.	Using the extended Euclidean algorithm, find the gcd (291,42) and the value of s and t.	(2)
6.	Use the rail fence technique to encipher the message “Meet me at the park today evening” with depth 2.	(2)
7.	Encrypt the plaintext “ATTACKATDAWN” using the Vigenère cipher with the repeating keyword “LEMON.”	(2)
8.	Using the Euclidean algorithm, find the greatest common divisor of the following pairs of integers. a. 88 and 220 b. 300 and 42	(2)
9.	Distinguish between cryptography and steganography.	(2)
10.	Decrypt the ciphertext “WKHHDJOHKDVODQGHG” using the Caesar cipher.	(2)
11.	Distinguish between a substitution cipher and a transposition cipher.	(2)
12.	Decrypt the message “ZEBBW” using the Affine cipher with key (7, 3).	(2)
13.	Encrypt the message “this is an exercise” using the Additive cipher with a key of 20.	(2)
14.	Why is the Vigenere cipher more secure than the Caesar cipher?	(2)
15.	Encrypt a message “SECURITY” using a columnar transposition cipher with the key 4312.	(2)
16.	Find the modulo inverse of 3 modulo 11 using the Extended Euclidean algorithm, and find GCD and S.	(2)
17.	Encrypt “INFORMATION TECHNOLOGY” using the Rail Fence technique with 3 rails.	(2)
18.	Encrypt the message “CRYPTOGRAPHY” using Columnar Transposition with the key “4312567”.	(2)
19.	A message “SECURITY” is to be encrypted using the Caesar cipher with a shift of 4. Construct the encrypted text and retrieve the original plaintext from the ciphertext.	(2)
20.	Use the Vigenere Cipher to encrypt the message “She is listening” with the keyword “PASCAL.”	(2)

PART-B
(16 Marks)

Q.No.			Marks
1.	Use a Hill cipher to encrypt the message “we live in an insecure world.” Use the key $\begin{pmatrix} 3 & 2 \\ 5 & 7 \end{pmatrix}$ and show your calculation in detail.		(16)
2.	Use a brute-force attack to decipher the following message. Assume that you know it is an affine cipher and that the plaintext “ab” is enciphered to “GL.” XPALASXYFGFUKPXUSOGEUTKCDGFXANMGNVS		(16)
3.	(i)	Use the Playfair cipher to encipher the message “The key is hidden under the door pad.” The secret key can be made by filling the first and part of the second row with the word “GUIDANCE” and filling the rest of the matrix with the rest of the alphabet.	(10)

<table>
<tr><td></td><td>(ii)</td><td>Use the Vigenere cipher with keyword “HEALTH” to encipher the message “Life is full of surprises”.</td><td>(6)</td></tr>
<tr><td>4.</td><td colspan="2">Using this Playfair matrix:
<table><tr><td>J/K</td><td>C</td><td>D</td><td>E</td><td>F</td></tr><tr><td>U</td><td>N</td><td>P</td><td>Q</td><td>S</td></tr><tr><td>Z</td><td>V</td><td>W</td><td>X</td><td>Y</td></tr><tr><td>R</td><td>A</td><td>L</td><td>G</td><td>O</td></tr><tr><td>B</td><td>I</td><td>T</td><td>H</td><td>M</td></tr></table>Encrypt this message:
I only regret that I have but one life to give for my country.</td><td>(16)</td></tr>
<tr><td>5.</td><td colspan="2">Encrypt the message “this is an exercise” using the following ciphers. Ignore the space between words.
i. Additive cipher with key = 20
ii. Multiplicative cipher with key = 15
iii. Affine cipher with key (15, 20)</td><td>(16)</td></tr>
<tr><td>6.</td><td colspan="2">An intelligence team uses layered encryption to protect a critical message, “OPERATION SAFE.” They apply three encryption techniques in sequence Caesar cipher, Affine Cipher (a = 5, b = 8), and Columnar Transposition Cipher with key “3142”.
i. Encrypt the plaintext step-by-step using all three ciphers in the specified order. Show all intermediate outputs and calculations. (8 marks)
ii. Decrypt the final ciphertext to retrieve the original message. Ensure all reverse operations are shown clearly. (8 marks)</td><td>(16)</td></tr>
<tr><td>7.</td><td colspan="2">Use a brute-force attack to decipher the following message enciphered by Alice using an additive cipher. Suppose that Alice always uses a key that is close to her birthday, which is on the 13th of the month:
NCJAEZRCLASJLYODEPRLYZRCLASJLCPEHZDTOPDZQLNZTY</td><td>(16)</td></tr>
<tr><td>8.</td><td colspan="2">Combine Caesar Cipher and Vigenère Cipher: Encrypt “HELLOWORLD” first using Caesar cipher with shift 5, then encrypt the result with Vigenère cipher using key “KEY”. Show full transformation for both steps.</td><td>(16)</td></tr>
<tr><td>9.</td><td>i.</td><td>Eve secretly gets access to Alice’s computer and, using her cipher types “abcdefghij.” The screen shows “CABDEHFGIJ.” If Eve knows that Alice is using a keyed trans position cipher, answer the following questions:
a. What type of attack is Eve launching?
b. What is the size of the permutation key?</td><td>(8)</td></tr>
</table>

	ii.	Encrypt the message “the house is being sold tonight” using the Vigenere cipher with key “dollars.” Ignore the space between words. Decrypt the message to get the plaintext.	(8)
10.	Encrypt the message “meet me at the usual place at ten rather than eight o clock’’ using the Hill cipher with the key $\begin{pmatrix} 7 & 3 \\ 2 & 5 \end{pmatrix}$. Show your calculations and the result.		(16)

Course Outcome	**CO2:** implement symmetric and asymmetric cryptographic techniques to ensure data security.

PART-A (2 Marks)		
Q.No.		Marks
1.	Distinguish between a stream cipher and a block cipher.	(2)
2.	Give information about DES: i. block size ii. cipher key size iii. round-key size	(2)
3.	Why does the DES function need an expansion permutation?	(2)
4.	What is triple DES? What is triple DES with two keys? What is triple DES with three keys?	(2)
5.	List the block cipher modes of operation.	(2)
6.	List the parameters (block size, key size, and the number of rounds) for the three AES versions.	(2)
7.	Compare DES and AES. Which one is bit-oriented? Which one is byte-oriented?	(2)
8.	Which of the four transformations defined for AES changes the contents of bytes? Which one does not change the contents of the bytes?	(2)
9.	Define the Chinese remainder theorem.	(2)
10.	Distinguish between symmetric-key and asymmetric-key cryptosystems.	(2)
11.	Distinguish between public and private keys in an asymmetric-key cryptosystem.	(2)
12.	To understand the security of the RSA algorithm, find d if you know that e = *17* and n = *187*.	(2)
13.	Define elliptic curves and their applications.	(2)
14.	In RSA, given p = 19, q = 23, and e = 3, find n, $\phi(n)$, and d.	(2)
15.	What are three broad categories of applications of public-key cryptosystems?	(2)
16.	What requirements must a public-key cryptosystem fulfill to be a secure algorithm?	(2)

17.	List the different approaches to attack the RSA algorithm.	(2)
18.	What is the zero point of an elliptic curve?	(2)
19.	State the purpose of the initial permutation in DES.	(2)
20.	Why is DES vulnerable to brute-force attacks today?	(2)

PART-B
(16 Marks)

Q.No.			Marks
1.	(i)	Differentiate between stream ciphers and block ciphers with suitable examples.	(4)
	(ii)	Discuss the modes of operation of block ciphers (ECB, CBC, CFB, OFB, CTR) and analyze their strengths and weaknesses.	(12)
2.	Explain the working of the Data Encryption Standard (DES) algorithm in detail with a neat diagram. Why is DES considered insecure today?		(16)
3.	A plaintext message is encrypted using Triple DES in EDE (Encrypt–Decrypt–Encrypt) mode with three different keys. Demonstrate the encryption and decryption process with an example and discuss its security advantage over DES.		(16)
4.	Describe the AES encryption algorithm and illustrate the operations: SubBytes, ShiftRows, MixColumns, and AddRoundKey with examples.		(16)
5.	(i)	Explain the principle of Elliptic Curve Cryptography (ECC).	(8)
	(ii)	Compare ECC and RSA in terms of key size, performance, and security. Provide an example of ECC-based key exchange.	(8)
6.	Explain the RSA algorithm in detail with an example.		(16)
7.	A user selects two prime numbers $p = 11$ and $q = 17$ for RSA encryption. i. Compute n and Euler's Totient $\varphi(n)$. ii. Choose a public key $e = 7$ and compute the corresponding private key d. iii. Encrypt the message $M = 88$ using RSA. iv. Show the decryption process and recover the original message.		(16)
8.	(i)	Discuss the challenges of private key distribution in symmetric key cryptography.	(6)
	(ii)	Explain solutions such as Key Distribution Centers (KDC), Diffie-Hellman Key Exchange, and Public Key Infrastructure (PKI) with suitable diagrams.	(10)
9.	An organization uses RSA for digital signatures with parameters: $p = 13$, $q = 19$, and $e = 5$.		(16)

	i. Compute the modulus n and private key d. ii. A message $M = 9$ is to be signed by the sender. Generate the digital signature using RSA. iii. Show how the receiver verifies the signature to confirm the authenticity of the sender.	
10.	Find the value of x for the following sets of congruence using the Chinese remainder theorem. i. $x \equiv 2 \bmod 7$, and $x \equiv 3 \bmod 9$ ii. $x \equiv 4 \bmod 5$, and $x \equiv 10 \bmod 11$ iii. $x \equiv 7 \bmod 13$, and $x \equiv 11 \bmod 12$	(16)

Course Outcome	**CO3:** Apply cryptographic techniques to ensure data integrity, secure communication, and authentication in network systems.

PART-A (2 Marks)		
Q.No.		Marks
1.	What characteristics are needed in a secure hash function?	(2)
2.	What is the role of a compression function in a hash function?	(2)
3.	How safe is it to use a hash function without encryption in terms of integrity?	(2)
4.	What basic arithmetical and logical functions are used in SHA?	(2)
5.	Summarize the five main steps of SHA-512 and its main functions.	(2)
6.	What types of attacks are addressed by message authentication?	(2)
7.	What are some approaches to producing message authentication?	(2)
8.	When sending a message to B, A can use A's private key and B's public key to achieve both secrecy and authentication. Which key is used to achieve which goal?	(2)
9.	What is a message authentication code?	(2)
10.	Differentiate between a message authentication code and a one-way hash function.	(2)
11.	In what ways can a hash value be secured to provide message authentication?	(2)
12.	List two disputes that can arise in the context of message authentication.	(2)
13.	What are the properties a digital signature should have?	(2)
14.	What requirements should a digital signature scheme satisfy?	(2)
15.	What are some threats associated with a direct digital signature scheme?	(2)
16	What is the difference between a session key and a master key?	(2)

17.	What are the requirements for the use of a public-key certificate scheme?	(2)
18.	What is the purpose of the X.509 standard?	(2)
19.	Which approach to secure user authentication in a distributed environment does Kerberos support?	(2)
20.	In Kerberos, what is the purpose of the centralized authentication server?	(2)

PART-B (16 Marks)			
Q.No.			Marks
1.	Explain the design and working of the MD5 algorithm with a neat diagram, and discuss its limitations and reasons for being considered cryptographically weak today.		(16)
2.	Describe the SHA-512 algorithm in detail, explaining the message padding, block processing, and compression functions involved in generating the hash.		(16)
3.	(i)	Differentiate between HMAC and CMAC.	(4)
	(ii)	Explain the working of HMAC in detail with a block diagram.	(12)
4.	Using the RSA scheme, let p = 809, q = 751, and d = 23. Calculate the public key e. Then a. Sign and verify a message with M_1 = 100. Call the signature S_1. b. Sign and verify a message with M_2 = 50. Call the signature S_2. c. Show that if $M = M_1 \times M_2$ = 5000, then $S = S_1 \times S_2$.		(16)
5.	(i)	Using the ElGamal scheme, let p = 881 and d = 700. Find values for e_1 and e_2. Choose r = 17. Find the value of S_1 and S_2 if M = 400.	(8)
	(ii)	Using the Schnorr scheme, let q = 83, p = 997, and d = 23. Find values for e_1 and e_2. Choose r = 11. If M = 400 and $h(\dots)$ = 100, find the value of S_1, S_2, and V. Is $S_1 \equiv V (\text{mod } p)$?	(8)
6.	Alice and Bob use the Diffie–Hellman key exchange technique with a common prime q = 157 and a primitive root α = 5. a. If Alice has a private key X_A = 15, find her public key Y_A. b. If Bob has a private key X_B = 27, find his public key Y_B. c. What is the shared secret key between Alice and Bob?		(16)
7.	Alice and Bob use the Diffie–Hellman key exchange technique with a common prime q = 23 and a primitive root α = 5. a. If Bob has a public key Y_B = 10, what is Bob's private key X_B? b. If Alice has a public key Y_A = 8, what is the shared key K with Bob? c. Show that 5 is a primitive root of 23.		(16)

8.	Explain the process of digital signature generation and verification with an example.		(16)
9.	Explain the working of the Kerberos authentication system with a neat diagram and how it ensures secure authentication in a distributed environment.		(16)
10.	(i)	Describe the X.509 Authentication Service**,** the structure of X.509 certificates, and their role in authentication.	(8)
	(ii)	Explain Public Key Infrastructure (PKI) components (Certificate Authority, Registration Authority, Certificates, and Repositories) and how PKI supports secure communications over the Internet.	(8)

Course Outcome	**CO4 :** Utilize security techniques to detect and prevent network intrusions t provide a safe environment.

PART-A (2 Marks)		
Q.No.		Marks
1.	What is the primary purpose of TLS/SSL in network security?	(2)
2.	Distinguish between a session and a connection.	
3.	Mention any two advantages of using SSL/TLS in secure communications.	
4.	What does S/MIME stand for? List any two security services provided by S/MIME.	
5.	What type of encryption (symmetric/asymmetric) does S/MIME use for message confidentiality?	
6.	Distinguish between the two modes of IPSec.	
7.	Define Security Association (SA) and its purpose.	
8.	What are the types of secret key algorithms used in IPsec?	
9.	List three design goals for a firewall.	
10.	List four techniques used by firewalls to control access and enforce a security policy	
11.	What information is used by a typical packet filtering firewall?	
12.	What are the main logical components of an IDS?	
13.	List the main categories of malicious software.	
14.	What is a distributed denial-of-service system with an example?	
15.	What is the main purpose of a VPN?	
16.	Name two types of VPNs commonly used in organizations.	
17.	How does a VPN ensure the confidentiality of transmitted data?	

18.	How do Next-Generation Firewalls differ from traditional firewalls?	
19.	Name two capabilities of Next-Generation Firewalls in application-layer security.	
20.	What is the role of deep packet inspection (DPI) in Next-Generation Firewalls?	

PART-B (16 Marks)		
Q.No.		Marks
1.	A corporate website is repeatedly targeted by man-in-the-middle attacks despite using HTTPS. Discuss how TLS/SSL mechanisms can be enhanced to prevent such attacks.	(16)
2.	Compare and contrast SSL and TLS in terms of architecture, security features, and vulnerabilities. Illustrate your answer with a suitable diagram.	(16)
3.	Explain the design and working of S/MIME for secure email communication and discuss the roles of encryption, digital signatures, and certificate management.	(16)
4.	Describe the process of IPSec key exchange and Security Association (SA) establishment, and how IPSec ensures secure communication over insecure networks.	(16)
5.	Explain the concept of Virtual Private Networks (VPNs), types of VPNs (site-to-site and remote access), and how VPNs provide confidentiality, integrity, and authentication.	(16)
6.	A multinational organization has recently experienced multiple security breaches due to malware and unauthorized access to sensitive applications. The IT team plans to deploy a Next-Generation Firewall (NGFW) to enhance security. As a network security consultant, explain how the NGFW can help mitigate these security risks and discuss its features, such as application awareness, intrusion prevention, threat intelligence, and deep packet inspection (DPI).	(16)
7.	A company has multiple branch offices across different cities, and employees often work remotely. They need a secure communication solution to ensure the confidentiality and integrity of data between branches and for remote workers. As a network engineer, explain the differences between site-to-site VPN and remote-access VPN, and discuss how encryption, tunneling protocols, and authentication mechanisms ensure secure communication.	(16)

8.	Explain the architecture and working of Network Intrusion Detection Systems (NIDS) and Network Intrusion Prevention Systems (NIPS) with a neat diagram.		(16)
9.	Discuss signature-based and anomaly-based detection techniques with examples.		(16)
10.	Explain the process of IPSec key exchange and Security Association (SA) establishment, and how IPSec ensures secure communication over insecure networks.		(16)
11.	(i)	Explain the features and capabilities of a Next-Generation Firewall (NGFW).	(8)
	(ii)	Discuss its differences with traditional firewalls and its role in application-layer security, intrusion prevention, and threat intelligence.	(8)

Course Outcome	**CO5 :** Identify and mitigate cyber threats through incident response strateg and secure coding practices.

PART-A (2 Marks)		
Q.No.		Marks
1.	Differentiate between phishing and spear-phishing attacks.	(2)
2.	A user notices that their personal data, browsing history, and login credentials have been leaked without their knowledge. In what way can spyware compromise user privacy in such a situation?	(2)
3.	Identify a secure coding practice to avoid injection flaws.	(2)
4.	Calculate the entropy of a password with 12 characters, each chosen from 95 printable ASCII characters.	(2)
5.	Difference between cross-site scripting (XSS) and cross-site request forgery (CSRF).	(2)
6.	A web application is vulnerable to an XSS attack. What quick coding practice can help mitigate this vulnerability?	(2)
7.	Write any four methods to reduce the risk of phishing attacks.	(2)
8.	Mention any four impacts of cross-site scripting (XSS).	(2)
9.	How does incident response triage prioritize security incidents?	(2)
10.	What is the role of encryption in mitigating data exfiltration threats?	(2)
11.	Mention two logging practices critical for effective incident response.	(2)
12.	Mention the purpose of incident response in cybersecurity.	(2)

13.	Ransomware spreads by sending itself to 5 new systems every 30 minutes. If it starts from 1 infected machine, calculate the number of infected systems after 4 hours.	(2)
14.	Identify any four common web vulnerabilities apart from OWASP Top 10.	(2)
15.	Differentiate between internal threats and external threats in cybersecurity.	(2)
16.	Mention any four uses of threat intelligence in cybersecurity.	(2)
17.	What are the stages of a ransomware attack lifecycle?	(2)
18.	How does Cross-Site Scripting (XSS) enable attackers to steal cookies?	(2)
19.	How do exploit kits facilitate web-based attacks?	(2)
20.	Identify the role of threat intelligence in proactive incident response.	(2)

PART-B (16 Marks)			
Q.No.			Marks
1.	Cross-Site Request Forgery (CSRF) vulnerabilities are found in an online service. Evaluate the attack mechanism and the preventive coding measures.		(16)
2.	Describe the risks posed by Cross-Site Scripting (XSS) in a web application and the coding practices to detect, remediate, and prevent this vulnerability.		(16)
3.			
	(i)	Describe the differences between Network Intrusion Detection Systems (NIDS) and Network Intrusion Prevention Systems (NIPS) and their roles in managing cyber threats.	(8)
	(ii)	Describe the different types of phishing attacks and outline effective incident response measures to handle a successful phishing incident.	(8)
4.	A cyberattack campaign uses multiple exploit kits to compromise vulnerable browsers. Discuss how exploit kits enable web-based attacks and propose defense mechanisms.		(16)
5.	An employee falls victim to a phishing email that compromises company credentials. Explain the incident response steps and how to strengthen user awareness and technical defenses.		(16)
6.	Your network is under a Distributed Denial-of-Service (DDoS) attack. Discuss the step-by-step incident response plan to mitigate the attack and restore service availability.		(16)

7.	A multinational firm wants to establish a robust incident response plan for future cyber threats. Evaluate the key phases of incident response and their role in mitigating damage during attacks.		(16)
8.	A hospital IT system is crippled by a ransomware attack demanding payment in cryptocurrency. Analyze the stages of a ransomware attack lifecycle and recommend incident response strategies.		(16)
9.	A critical system experiences unauthorized access from multiple unknown IP addresses. Describe your immediate and long-term incident response actions to secure the system and prevent future breaches.		(16)
10.			
	(i)	Multiple failed login attempts are observed targeting an admin account. Describe how you would investigate and defend against potential brute force attacks.	(8)
	(ii)	Your security team detects keylogger installations on several corporate devices. Outline detection methods and prevention strategies for keyloggers.	(8)